part of the researcher/experimenter as they rescue them from repetitive tasks like circuiting the basic hardware every time.

It becomes a cumbersome task to work on a microcontroller-based research project when in order to use a microcontroller you need to set up hardware circuits every time. This can get pretty frustrating considering most of the parts of a basic setup are always the same.

So, what if all the required hardware were already set? This would make experimentation and prototyping a lot easier and interesting. This is what a development board is made to provide.

Some of the basic components common to all development boards are as follows:

- Power Circuit (it is typically a circuit that runs on either 9V or 12V potential)
- Program Interface (to program the microcontroller using a computer)
- Basic input components like buttons
- Basic output components such as LEDs
- I/O pins

If you want to build a basic board using a general-purpose PCB or a breadboard, you can't always be sure if all the connections are good to work or not. 9 out of 10 times there will be something that doesn't work and you will unable to pinpoint for a while if it is due to setup or the nature of development you are carrying out. This wastes a lot of time. Also, sourcing components and assembly plus ensuring the compatibility of your set up with the components you are going to use are time-consuming in itself.

Thus, it is a lot better to use a dedicated development board as per your requirements with a ready interface and immediately start with your project. These dev boards can be used in a wide variety of projects, so if you are in the industry, it is all but a one-time investment.

Popular Development Boards that are the best in class

Now that we know about development boards and their use in the industries, research, and projects, let's focus on knowing in detail some of the popular development boards that are best in class. This section of the chapter will guide the readers in choosing the right development board for their project or research.

1. Arduino Uno R3

The best part about Arduino is that it creates perhaps the most user-